MethylPhys CPG - Operations Manual, chain v3

Development build, not commissioned; not a diagnostic test. Generated 2026-10-15 by manual/build_manual_v3.py from the files named at the start of each part. The procedure is sop/MethylPhys_CPG_SOP_v3.md.

Part 1. Operations - chain v3, neutrophils

Source: `manual/OM_v3_neutrophil_chain.md`

Operations Manual — chain v3, neutrophils (operator chapter)

Build: v3 (2026-10-01). Met-A on isolated neutrophils COMMISSIONED 2026-10-09; Met-A C-score on whole blood COMMISSIONED 2026-10-15; every other stage DEVELOPMENT - not commissioned; this chapter re-checked against the code on 2026-10-03 and updated for development round 2 on 2026-10-04; Stage T updated on 2026-10-04 for self-tare II then the median tare, adopted by the author (`boxruns/run1/JOBS.md` job A; `doors/DEV_SELFTARE_02.md`; development log 2026-10-04, DEV-PAIRED-01). Until then Stage T was the median tare alone (DEV-TARE-02, 2026-10-02). Self-tare II was wired into Stage T on 2026-10-04 (development log DEV-SELFTARE-03). The procedure is in `sop/MethylPhys_CPG_SOP_v3.md`; this chapter covers running it and reading the output. The PDF manual in this folder (`MethylPhys_CPG_Operations_Manual.pdf`) is built from this chapter, the generated chain sequence and the toolkit list by `build_manual_v3.py`. The class-floor engine (v2) and its manual were retired on 2026-10-03 and are archived privately. Paths are relative to `Biological_Physics/MethylPhys/`.

Before a run

1. Use EPIC v1 IDAT pairs (Grn and Red) of whole blood or isolated / sorted / purified neutrophils. Any other specimen (PBMC, other sorted fractions, bone marrow, cell lines, tissue, unspecified) is refused at intake with a report. Sex and age are optional: give them when known and they are recorded; with a declared sex Stage 0.8 compares it with the array.
2. Put at least 3 healthy reference specimens of the same specimen type on the same slide (else the same batch), run the same way, so each reading can be tared. This applies to whole blood and to isolated neutrophils. Stage T is self-tare II (each array tared on its own fixed sites; no references needed), then the median tare against these references.
3. Use Python 3.11 with the versions in `chain/requirements.txt` (methylprep 1.7.1, numpy 1.26.4, pandas 1.5.3, scipy 1.17.1; reportlab for the manual): `pip install -r chain/requirements.txt`.
4. Stage 1 needs the Illumina manifest. methylprep downloads it on first use into `$HOME/.methylprep_manifest_files/` (`HOME` must be writable; network access to `array-manifest-files.s3.amazonaws.com`); offline, place the files there by hand (`doors/RUNBOOK.md` section 1). In a batch, run the first array alone so the download is not raced.
5. The frozen inputs are in the repository under `chain/Runtime Matrices/` (`Met_A_Floors/`, `IAM_A_Positions/`, `Intake/`); their hashes are in `chain/FROZEN_INPUTS_v3.json`.

Run

From `chain/MethylPhys_Interface/` (run on the box: Stage 1 needs the manifest):

```
python run_sample.py --grn <Grn> --red <Red> --engine v3 --specimen "whole blood" --array-type EPIC_v1 \
  --id <id> --out <id>.html          # add --sex <F|M> --age <years> when known
```

- Isolated neutrophils: `--specimen "isolated neutrophils"`.
- Pass 2 (the tare): run each specimen again, whole blood and isolated alike, with `--slide-ref-A a1,a2,a3`, where those are the references' A values from pass 1 (`met_a.A`, self-tared, before the median tare), or with

`--slide-ref-table refs.csv` (column A, optional column `id`; a row whose `id` is the specimen's own `--id` is left out). Fewer than 3 references: the median tare does not run and the reading stays untared.

- Self-tare II (Stage T step 1, adopted 2026-10-04): always runs (wired into Stage T on 2026-10-04); no flag is needed. It maps each probe design onto the reference arrays' scale by this array's own low and high fixed-site anchors before Met-A is read (`tare.selftare_ii`). The printed A_rel is the median tare of the self-tared A. The noise index reads the betas before self-tare II. `--dev-selftare-ii` is a no-op alias, kept so recorded commands still run.
- A beta table already calibrated by this chain's Stage 1 (two-column CSV `cpg_id,beta`; Stage 0 does not run): `python run_sample.py --betas <table>.csv --specimen "whole blood" --id <id> --out <id>.html`.
- Second draw of the same person (stage 12b): run the first draw with `--save-betas <id>_betas.parquet --patient-id <hash>`; run the second with the same `--patient-id` plus `--prior-betas <id>_betas.parquet --prior-bundle <id>_bundle.json`.
- Identifiers: the bundle and the ledger carry the sha256 hash of `--id`; the report keeps the id you typed.
- Development flags (DEVELOPMENT - not commissioned; never part of the reading; `--dev-selftare-ii` is a no-op alias since the adopted Stage T step 1 was wired in on 2026-10-04): `--dev-selftare-ii --dev-direction --dev-trace --dev-foreign --dev-brightness` (no extra input); `--dev-nilc --dev-atlas-e --dev-percell-b --dev-sky` with `--atlas-v2 <IAMAtlas_v2.parquet>` (`--dev-sky` needs healpy); `--dev-epic-v2 --sesame-rscript <Rscript>` for an EPIC v2 IDAT pair. Each writes `development.<stage>` into the bundle and a report section.
- Sequencing (Stages Q0 and Q, IAM-A): `python run_sample.py --pat <file>.pat.gz --specimen "isolated neutrophils" --id <id> --out <id>.html` (add `--alignment-qc <qc>.json` when the alignment step wrote one); read the whole file - a `--pat-max-bytes` cut is refused, or `python run_sample.py --site-table <sites>.csv --seq-pipeline loyfer_pat_v1 --id <id> --out <id>.html`.

Output: `<id>.html`, `<id>_bundle.json` beside it, and one row appended to `evidence_ledger.jsonl` in the same folder (`--ledger` to change; `--no-bundle` writes neither the bundle nor the row). Exit 0 when a report is written (a refusal still writes one); exit 2 on a Stage 0 QUARANTINE, with no report, no bundle and no ledger row (a command-line error also exits 2).

Batch runners (written for the compute box; they read box paths and roster files and are not general tools): `chain_tests/run_chain_acceptance.py` (pass 1 every specimen; pass 2 re-runs each whole-blood specimen with the other specimens of its group as references; isolated specimens are not re-run) and `doors/data/DEV_REPL_V3_01_run/run_proc_repl_v3_01.py` (pass 2 with `--slide-ref-table` built from the other arrays on the same slide, else the batch; exact commands in `COMMANDS.md` beside it).

Checks: `python chain/release_check_v3.py` (or `python kit/release_check.py`) - exit 0 only when every check passes (F1, F1b, S1-S4, E1-E10, M1). The IDAT checks E1-E3 and E6 need the manifest (run on the box). Manual: `python manual/build_manual_v3.py` rebuilds the PDF.

Reading the report

field	meaning
Refused (specimen)	`SPECIMEN_REFUSED`: the specimen has no reference in chain v3; the refusal names it. No reading, Stage 1 does not run
Refused	the platform check: not an EPIC v1 vector (array type, EPIC v2 probe names, or 700,000 probes or fewer); an EPIC v2 IDAT pair is refused at intake, before Stage 1. No reading
Commissioning box (Stage 5)	green box: Met-A COMMISSIONED 2026-10-09 (isolated neutrophils); the detection limit (2 %: the smallest loss that reliably reads outside Normal) and this run's own statistical limit (2 × reference spread). Present only when Met-A was read on isolated neutrophils; a whole-blood reading is labelled DEVELOPMENT, with its 5 % detection limit

field	meaning
Stage 0 intake	verdict PROCEED / PROCEED_WITH_PENALTY / QUARANTINE (QUARANTINE produces no report), call rate, flags; `not run` for `--betas`, `--pat`, `--site-table` or `--no-intake`
Stage 1	poobah detection, call rate and controls: recorded, not gated
Stage A composition	the 8 blood groups (groups at 1 % or more are listed). Whole blood is read when neutrophils are ≥ 20 % and ≥ 867 of the 963 markers are measured
Stage M Met-A	isolated cells: A against the healthy reference, drawn on the gauge as untared until tared. Whole blood untared: A is a number only, no gauge position. Shift per 1 % loss of the neutrophil pattern
Stage T tare	adopted 2026-10-04: self-tare II (β′ = Lr + (β – L)(Ur – Lr)/(U – L) per probe design, from this array's own fixed-site anchors L, U and the reference arrays' Lr, Ur; nothing fitted), then the median tare. Printed: A_rel = A ÷ median of the references (self-tared A; self-tare II wired in on 2026-10-04, record `tare.selftare_ii`); number of references and their median; reference spread; detection limit (% loss of the pattern) = 2 × spread ÷ shift per 1 % loss. **Normal = 0.95–1.05**
Noise index N	mean H(β) on the 48,528 noise sites. Above 0.149 on an untared reading: state `withheld`, A printed as a number. Fewer than 90 % of the noise sites measured: N is not formed and the state is withheld, tared or not, with the counts and the reason
Methylated sites mean β	below 0.5: past the entropy ceiling; read β, not A
Stage MC C-score	genomic clustering of the departures in blocks of 10 sites (healthy = 1), with the healthy held-out range; C_rel = C tared against the same-run healthy references when ≥ 3 are given. **COMMISSIONED 2026-10-15 on whole blood**: one-sided call (above 1.152: clustered departure; below 0.877: less clustered than healthy, not a departure), detection limit C_rel 1.5 at 5 % of sites clustered, and the commissioning record printed; on isolated neutrophils labelled development
Stage Q0 intake	sequencing only: each intake check with its value and result (pass / REFUSED / recorded / not measured)
Stage Q IAM-A	sequencing only: IAM-A, copy error eps, position P, eps0, the two halves, opportunities; the IAM-A C-score (development: independent copy errors give 1; band not set); with `--iama-ref-table` (≥ 3 same-run healthy references, same kit and laboratory) A_rel and its state, else 'untared' with the reason
Development stages	only with a development flag: one row per flagged stage, labelled DEVELOPMENT - not commissioned
Withheld	what the build does not print, and why
Stage 12b difference map	only with `--prior-betas`/`--prior-bundle`: per-address difference to an earlier draw of the same person, or the refusal naming what differs (identifier hash, array type, pipeline)
Red flags	STOP / WITHHELD / CAUTION / NOTE, from the bundle; also written to the bundle as `red_flags`
Safeguards	rendered-claim scan (no disease, cohort or population words in the prose), formula self-test, anchors (Normal band, floor, C baseline, N_max read from the frozen files), deconvolver conformance, atlas separability (NOT_RUN until stage 3 is wired)
Troubleshooting, Integrity, Chain file inventory, Run it yourself	what to do for each condition on this specimen; IDAT hashes and chain commit; every frozen input and module with its hash; the exact command
The chain, Toolkit stages	SOP 2b stage table; each toolkit stage PASS / FAIL / REFUSED / NOT_RUN / NOT_BUILT on this specimen

Faults

symptom	cause	action
QUARANTINE_INCOMPLETE_MANIFEST	array type not declared and not readable from the IDAT header (sex and age are optional since 2026-10-04)	supply `--array-type`
`SPECIMEN_REFUSED`	the specimen is not whole blood or isolated / sorted / purified neutrophils	none: that specimen needs its own reference first

symptom	cause	action
`withheld: only <n> of the 48,528 noise sites were measured`	fewer than 90 % of the noise sites passed Stage 1 detection	re-hybridise or check the array's signal; the state cannot be shown without the array's own noise
ENVIRONMENT_MISSING_MANIFEST (exit 3; before 2026-10-03 this showed as QUARANTINE_CORRUPT_IDAT) with PermissionError or a connection error on `.methylprep_manifest_files`	the manifest is not in its cache and could not be downloaded; the IDAT is not at fault and is not judged	make `HOME` writable and allow the download, or place the manifest files by hand; re-run
Stage 0 QUARANTINE, other hard failures (`call_rate`, `detection`, `ctrl_qc`, `sex`, `integrity`, `hm450_coverage`, `intake_deferred:...`)	the array failed intake	none: no reading by rule; the flags name the check
`neutrophil fraction <f> < 0.2: fraction reported, A withheld`	neutrophils < 20 % of the whole blood	none: the fraction is reported and A is withheld by rule
`only <n> of 963 composition markers measured` / `only <n> of 6000 ... sites measured`	fewer than 90 % measured	none: A withheld
`untared: <n> same-run reference arrays (>= 3 required)`	fewer than 3 references supplied	run the references, then pass them with `--slide-ref-A` or `--slide-ref-table`
`tare.selftare_ii` NOT_RUN, `dev_selftare_typeII_EPIC_v1.json not found` (the reading then uses β without self-tare II)	the self-tare II file is missing from `chain/Runtime Matrices/Development/`	restore it from the repository
`withheld: noise index <N> > 0.149 and no same-run tare`	the array is noisier than the reference arrays; the gauge is not drawn	tare it against same-run references
refusal `... (450K or incomplete vector) ... (450K neutrophil floor pending)`	fewer than 700,001 probes after Stage 1 (a 450K table, or an EPIC array that lost probes at detection)	none: v3 reads EPIC v1 only. A 450K IDAT pair never gets here: it quarantines at Stage 0 (0.1 array-type mismatch when `--array-type EPIC_v1` is given, else 0.7b coverage)
refusal `... (no frozen neutrophil floor for this platform)`	declared array type is not EPIC_v1, or EPIC v2 probe names	none: no floor for that platform
Stage Q0 `QUARANTINE_UNREADABLE_PAT` / `GENOME_BUILD_MISMATCH` / `QUARANTINE_NOT_WHOLE_GENOME` / `SPECIMEN_REFUSED`	a cut or corrupt file; a file on another genome build (e.g. hg38); a targeted or partial file; whole blood or another cell	download the file again; convert with the loyfer_pat_v1 pipeline on hg19; read a whole-genome file; IAM-A reads purified neutrophils only
IAM-A refusal `partial file`	the reading was cut with `--pat-max-bytes`	read the whole file (P is measured on whole files)
IAM-A refusal `position for neutrophils was measured on loyfer_pat_v1, not <pipeline>` / `too few opportunities`	another read-level pipeline, or fewer than 100,000 opportunities	none: P is valid only for its own pipeline

What each instrument can and cannot see (2026-10-11; record: doors/DEV_DIRECTION_AUDIT_01.md)

	Met-A (arrays)	IAM-A (sequencing)
What it reads	fixed identity sites, both kinds: sites that should be methylated and sites that should not	the cost of holding marks: copy error on the molecules a cell keeps methylated
Sees loss	yes (methylated identity sites)	yes (loss is copy error)
Sees gain	yes (unmethylated identity sites), about as strongly as loss	only as a mixture: gained molecules enter Stage Q and move IAM-A either way
Blind spot	only the sites it was built on; nothing about copy fidelity	which direction moved, unless held and gained territory are read separately

What IAM-A's copy error means: an isolated error in the held pattern, set by the balance of restore and loss (ε = f/(f+g)), whether the cell divides or not; division sets only how fast the balance is reached (SOP §1; book Sec. "The held state is a balance of restore and loss").

Rules that follow:

- Every new instrument, meter or sealed test is shown BOTH directions in its synthetic test (a planted loss and a planted gain), and its record states which it cannot see.
- A fingerprint is read in both directions: Arm B (loss) and Arm G (gain).
- The IAM-A C-score is a whole-genome reading. It is not read on RRBS: healthy RRBS files read 1.6 to 42 against its derived null of 1.
- Before any statistic is read on a new platform, its derived null is checked on healthy files of that platform.

Part 2. The chain, step by step

Source: `doors/CHAIN_SEQUENCE.md`

The chain, step by step - derived from the code

Generated by `chain/build_chain_sequence.py` from the AST of `run_sample.py`, `conductor_v3.py` and `stage_q_iam_a.py`. Do not edit by hand: re-run the generator. Every step below is a call the code actually makes, in the order it makes it. Chain v3 is the only engine; the class-floor engine (v2) was retired on 2026-10-03 and is archived privately.

The live path (chain v3) - 29 steps, in order

#	step	implemented in	what it does
1	`specimen_refusal`	`stage_0_intake.py`	None when intake accepts the specimen (ACCEPTED_SPECIMENS); otherwise the plain refusal text naming the specimen (decision L)
2	`step_0_1_idat_arrival`	`stage_0_intake.py`	SOP section 0.1_idat_arrival
3	`step_0_2_manifest_creation`	`stage_0_intake.py`	SOP section 0.2_manifest_creation
4	`step_0_3_integrity_hash`	`stage_0_intake.py`	SOP section 0.3_integrity_hash
5	`step_0_4_control_probe_validation`	`stage_0_intake.py`	SOP section 0.4_control_probe_validation
6	`step_0_5_detection_pvalue_qc`	`stage_0_intake.py`	SOP section 0.5_detection_pvalue_qc
7	`step_0_6_bead_count_qc`	`stage_0_intake.py`	SOP section 0.6_bead_count_qc
8	`step_0_7_call_rate`	`stage_0_intake.py`	SOP section 0.7_call_rate
9	`step_0_7b_platform_coverage`	`stage_0_intake.py`	SOP section 0.7b_platform_coverage
10	`step_0_8_sex_check`	`stage_0_intake.py`	SOP section 0.8_sex_check
11	`step_0_9_decision_gate`	`stage_0_intake.py`	SOP section 0.9_decision_gate
12	`Stage 1 - IDAT calibration`	`stage_1_idat_calibration.py`	Stage 1 - IDAT calibration to beta (SOP Stage 1, steps 1.1-1.2 + 1.5) _called by run_sample.py when given --grn/--red; skipped when given --betas_
13	`platform_refusal`	`conductor_v3.py`	None when the specimen is EPIC v1; otherwise the refusal text
14	`stage_t_selftare_ii`	`conductor_v3.py`	Stage T step 1, self-tare II (adopted 2026-10-04, DEV-SELFTARE-02): per probe design, beta' = Lr + (beta - L)(Ur - Lr)/(U - L) from this
15	`stage_m_isolated`	`conductor_v3.py`	Isolated / sorted neutrophils against their healthy reference. The healthy-reference state is kept as state_own_floor; the reading's state is
16	`stage_a_composition`	`conductor_v3.py`	EPIC blood composition (blood_composition_EPIC_v1): 8 groups from Salas purified EPIC cells; markers exclude the neutrophil sites;
17	`stage_m_blood`	`conductor_v3.py`	Whole blood: Met-A = mean H(beta) / mean H(e) at the neutrophil sites, e = sum_g f_g mu_g (EPIC purified group profiles)

#	step	implemented in	what it does
18	`noise_index`	`conductor_v3.py`	N = mean H(beta) over the noise sites measured on this array; None when fewer than MIN_NOISE_FRACTION are measured _stage 9: the array's noise index N over the noise sites_
19	`stage_t_tare`	`conductor_v3.py`	Same-run tare against >= MIN_REFS healthy references of the same specimen type run the same way (same slide, else same batch)
20	`noise_gate`	`conductor_v3.py`	Runtime Matrices/Met_A_Floors/noise_gate_EPIC_v1.json: N_max = top of the reference arrays' noise range (DEV-NOISE-01) _stage 9: N_max read here; run_neutrophil withholds the gauge state when N > N_max and the reading is untared_
21	`stage_mc_cscore`	`conductor_v3.py`	Met-A C-score (stage 6): clustering of the residual z map in genomic order - variance of the means of blocks of clustering_block
22	`stage_t_cscore`	`conductor_v3.py`	Same-run tare of the Met-A C-score, the same rule as the median tare of A (DEV-CSCORE-TARE-01): C_rel = C / median(C of >= MIN_REFS
23	`stage_q0_intake.intake`	`stage_q0_intake.py`	Stage Q0 on one .pat(.gz). alignment_qc: dict (or JSON path) from the alignment step with 'conversion_rate' and/or _runs when run_sample.py is given --pat, before Stage Q_
24	`stage_q_iam_a.pat_site_table`	`stage_q_iam_a.py`	Per-site table (pos, opp_A, err_A, opp_B, err_B) from one .pat(.gz) file, pipeline loyfer_pat_v1. pos = 'chrom:CpG index' _runs when run_sample.py is given --pat or --site-table_
25	`stage_q_iam_a.read`	`stage_q_iam_a.py`	site_table: per-site table (see module doc). cell: e.g. 'neutrophils'. pipeline: the read-level pipeline that produced the table _runs when run_sample.py is given --pat or --site-table_
26	`stage_q_iam_a.cscore`	`stage_q_iam_a.py`	IAM-A C-score for the pooled table (opp = opp_A + opp_B) and for each half. Returns {C, n_blocks, se_null, halves: {...}} _runs when run_sample.py is given --pat or --site-table_
27	`serial_mode.check_same_person`	`serial_mode.py`	Refuse unless patient hash, array type and pipeline match. Returns (ok, reason) _stage 12b: runs when run_sample.py is given --prior-betas and --prior-bundle_
28	`serial_mode.delta_sky`	`serial_mode.py`	beta_now - beta_prior per address on the addresses both draws measured. No expectation, no sigma, no composition _stage 12b: runs when run_sample.py is given --prior-betas and --prior-bundle_
29	`Report`	`MethylPhys_Interface/report_v3.py`	one self-contained HTML page plus the JSON bundle

Stage modules in chain/ that the live path does not call

module	what it implements	status
`stage_2c_trace_detection.py`	Stage 2c - is there any evidence of a trace class in this specimen?	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)
`stage_4_6_patient_cmb.py`	Stage 4.6 — the patient's sky (PROC-CMB-01, 2026-09-21). Replaces the retired patient_brightness_comparison.py	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)
`stage_a_composition_v2.py`	Stage A - composition, atlas v2 solver (deconv_v2). Not called by chain v3 (conductor_v3.stage_a_composition uses blood_composition_EPIC_v1.json). Frozen settings (commissioned as PROC-DECONV-V2-01; specimen cell sets 2026-09-30)	module present; neither run_sample.py nor conductor_v3.py calls it (toolkit, chain/TOOLKIT.md)

Behind development flags (DEVELOPMENT - not commissioned; never part of a reading)

flag	step	implemented in	what it does
`--dev-direction`	`dev_stages.direction_record`	`dev_stages.py`	Signed move toward or away from 0.5 at the neutrophil identity sites against the cell's own reference, tared on same-run references (DEV-DIRECTION-02)
`--dev-trace`	`dev_stages.trace_cell`	`dev_stages.py`	3b: another blood group in an isolated-neutrophil specimen, called above 3 x this array's own standard error (DEV-TOOLKIT-ADDED-02)
`--dev-foreign`	`dev_stages.foreign_cell`	`dev_stages.py`	3c: a non-blood template in whole blood by NNLS with the blood groups, called above 3 x this array's own standard error (DEV-TOOLKIT-ADDED-02)
`--dev-brightness`	`dev_stages.brightness`	`dev_stages.py`	11b: 95 % interval on Met-A from the v3 per-site uncertainty (this array's own fixed-site noise), by Monte Carlo (DEV-TOOLKIT-ADDED-02)
`--dev-nilc`	`dev_stages.nilc_e`	`dev_stages.py`	Stage 4 NILC-e composition on the atlas_e templates (DEV-NILC-01 method), 8-group sums and the noise per fraction
`--dev-atlas-e`	`dev_stages.atlas_e`	`dev_stages.py`	Stage 3 atlas_e composition (12 array-measured circulating atlas v2 cells, DEV-ATLAS-EPIC-02 method), 8-group sums
`--dev-percell-b`	`dev_stages.percell_b`	`dev_stages.py`	Stage 5 B-cell Met-A against the composition-matched expectation on the development B-cell floor (DEV-PERCELL-01 method)
`--dev-sky`	`dev_stages.sky`	`dev_stages.py`	Residual sky with the within-chromosome block-shuffle null and the look-elsewhere statistic by simulation (DEV-SKY-02)
`--dev-epic-v2`	`dev_stages.epicv2_calibrate`	`dev_stages.py`	EPIC v2 IDAT pair -> beta Series on EPIC v1 probe names (replicate v2 probes of one CpG averaged). SeSAMe openSesame(prep), pOOBAH-masked

The toolkit (chain/TOOLKIT.md) - built; each stage wired only after its check passed

stage (SOP 2b)	name	module	status
3	Atlas deconvolution	`chain/deconv_v2.py`	toolkit: not yet wired into chain v3
3b	Trace-cell detection	`chain/dev_stages.py`	toolkit: not yet wired into chain v3
3c	Foreign-cell detection	`chain/dev_stages.py`	toolkit: not yet wired into chain v3
4	NILC component separation	`chain/dev_stages.py`	toolkit: not yet wired into chain v3
4	NILC helper: lineage splitter	`chain/Lineage_Splitter/lineage_splitter.py`	toolkit: not yet wired into chain v3
10	Directional decomposition	`chain/dev_stages.py`	toolkit: not yet wired into chain v3
11	Sky map	`chain/dev_stages.py`	toolkit: not yet wired into chain v3
11b	Surface brightness	`chain/dev_stages.py`	toolkit: not yet wired into chain v3
12	Sky statistics	`chain/sky_statistics.py`	toolkit: not yet wired into chain v3
12b	Difference map	`chain/serial_mode.py`	wired into chain v3 (optional input; see chain/TOOLKIT.md)
5	Met-A per cell (B cells)	`chain/dev_stages.py`	toolkit: not yet wired into chain v3
8	Self-tare on type II fixed sites	`chain/dev_stages.py`	wired into chain v3: always runs (Stage T step 1); see chain/TOOLKIT.md
1	EPIC v2 calibration	`chain/dev_stages.py`	toolkit: not yet wired into chain v3

Part 3. The toolkit

Source: `chain/TOOLKIT.md`

chain/TOOLKIT.md - the toolkit: built, kept, entering chain v3 one stage at a time (2026-10-03: 12b wired; 2026-10-04, development round 2: stages likely to work behind development flags)

Every module below is in the repository and importable. **None of it is part of a reading**; stage 12b (difference map) is wired as an optional second-draw comparison and changes no reading. Since 2026-10-04 (author decision N, `doors/DEV_FLAGS_01.md`) the stages likely to work run behind development flags of `run_sample.py` (`chain/dev_stages.py`): each writes `bundle["development"][<stage>]`, labelled DEVELOPMENT - not commissioned, and none changes the reading. Development round 2 records: `doors/DEV_ROUND2_REPORT.md`. Each enters chain v3 at commissioning, one stage at a time, after its own pre-registered check has passed on v3 and the outcome is recorded in `doors/` (order and checks: `../sop/MethylPhys_CPG_SOP_v3.md` section 2b). Every module carries the line *Toolkit: not yet wired into chain v3; enters the chain at commissioning with its own pre-registered check.* Stage numbers and names are those of SOP v3 section 2b. Paths are relative to `Biological_Physics/MethylPhys/`. The class-era engine these modules were first used with (chain v2) was retired on 2026-10-03 and is archived privately; nothing here imports it.

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status						
3	Atlas deconvolution	`chain/deconv_v2.py` (DeconvV2); wrapper `chain/stage_a_composition_v2.py`; atlas_e (12 array-measured circulating cells) `chain/dev_stages.py` (`atlas_e`) with `chain/dev_comp_methods.py`; development copies `tlas/v2/postbuild/scripts/deconv_v2.py`, `methylphys_v2.py`	one joint non-negative solve over the atlas v2 cell types, f >= 0, sum f = 1, inverse-variance weights iterated, markers chosen per cell against its nearest cells	the atlas v2 reference (parquet, built by `tlas/v2/scripts`; not stored in the repository), a beta vector on the Stage 1 scale	development flag `--dev-atlas-e` (2026-10-04). Checks: DEV-ATLAS-EPIC-02 held-out truth outside the bars; DEV-COMPOSITION-TRUTH-02 (GSE77797, another laboratory, 450K): every group within 0.03 except granulocytes (0.041)						

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status						
3b	Trace-cell detection	`chain/dev_stages.py` (`trace_cell`, round 2: another blood group in an isolated-neutrophil specimen, line = 3 x this array's own standard error); class-era `chain/stage_2c_trace_detection.py` with `chain/Runtime Matrices/trace_detection_panel_v1.json` kept out (line from other arrays)	weighted regression of the residual (beta - neutrophil profile) on each group's contrast at the 963 markers; weights from the v3 per-site uncertainty	a beta vector; `Runtime Matrices/Development/dev_selftare_typeII_EPIC_v1.json` (per-site uncertainty)	development flag `--dev-trace` (2026-10-04). DEV-TOOLKIT-ADDED-02: 82.5 % called at 5 % (bar 95 %); 97.9 % with pure spike material; 0/360 unspiked						

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status						
3c	Foreign-cell detection	`chain/dev_stages.py` (`foreign_cell`, round 2: NNLS on the 8 groups + a non-blood template, line = 3 x this array's own standard error); class-era `chain/toolkit_foreign_detection.py` with `detection_panel_v3.json` kept out (line from other arrays, class-era scale)	non-negative fit at the markers plus sites where the template differs from every blood group by >= 0.25	a beta vector; `RuntimeMatrices/Development/dev_foreign_placenta_EPIC_v1.json`	development flag `--dev-foreign` (2026-10-04). DEV-TOOLKIT-ADDED-02: 100 % called at 5 %, but also 100 % with nothing added (zero offset 0.023-0.048): needs a same-run zero						
4	NILC component separation	NILC-e `chain/dev_stages.py` (`nilc_e`) with `chain/dev_comp_methods.py`; class-era `chain/nilc_celltype_deconvolver.py` (`NILCCelltypeDeconvolver`) kept out	internal linear combination per cell type on the atlas_e templates, constrained, noise per fraction printed	the atlas v2 parquet (not in the repository), a beta vector	development flag `--dev-nilc` (2026-10-04). DEV-NILC-01 truth bars outside; DEV-COMPOSITION-TRUTH-02: granulocytes 0.061, CD4T 0.036 outside. The class-era module is kept out (every truth bar outside in DEV-NILC-01)						

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status						
4	NILC helper: lineage splitter	`chain/Lineage_Splitter/lineage_splitter.py` (`LineageSplitter`)	splits one compartment's mass among its member classes by a small NNLS on the contrast CpGs (abs(mean_a - mean_b) >= min_delta, weighted by the contrast), with a condition-number check; reports the compartment only when the split is ill-conditioned	atlas CSV, a beta vector, the compartment fraction and its members	helper of stage 4; not wired (stage 4 check failed)						
10	Directional decomposition	`chain/dev_stages.py` (`direction`, `direction_record`, round 2, physics only); class-era `chain/Runtime Matrices/Directional Panel/bidirectional_decomposition.py` with `directional_panels_v1_0.json` kept out (z against other arrays, a disease sign)	D = mean over identity sites of \	r - 0.5\	- \	beta - 0.5\	against the cell's own reference; > 0 toward disorder; tared on same-run references (direction when \	D_rel\	> 2 x reference spread)	a beta vector, the frozen profiles	development flag `--dev-direction` (2026-10-04). DEV-DIRECTION-02: decitabine and NTX-30112/12 toward disorder; replicates 56/63 no direction (bar 95 %)

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status						
11	Sky map	`chain/dev_stages.py` (`residual_z`, `sky`); class-era `chain/stage_4_6_patient_cmb.py` (`patient_sky_sigma`, `render_plate`); mapping `chain/Runtime Matrices/Patient_CMB/iamatlas_cpg_to_healpix_nside128.npz` built by `chain/Runtime Matrices/Patient_CMB/build_healpix_mapping.py`	residual z = (beta - sum f mu) / sigma per CpG on HEALPix NSIDE 128 (RING) in genomic order	a beta vector, NNLS8 fractions, atlas v2 parent means and SDs, the mapping	development flag `--dev-sky` (2026-10-04, needs healpy). DEV-SKY-02: against the within-chromosome block-shuffle null, bands 2, 5, 6 inside 0.9-1.1; l 2-8 1.84, l 25-64 1.14, l 65-128 1.15						
11b	Surface brightness	`chain/dev_stages.py` (`brightness`, round 2: v3 per-site uncertainty); class-era `chain/toolkit_surface_brightness.py` (`attach_brightness_ci`) kept out (class archives)	95 % interval on Met-A by Monte Carlo of beta_i + N(0, s_i^2), s_i from this array's own fixed sites	a beta vector, `Runtime Matrices/Development/dev_selftare_typeII_EPIC_v1.json`	development flag `--dev-brightness` (2026-10-04). DEV-TOOLKIT-ADDED-02: half-width 0.003 against same-person differences 0.032; 3.8 % of pairs inside (bar 95 %)						

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status						
12	Sky statistics	`chain/sky_statistics.py` (`masked_spectrum`, `bandpowers`, `shuffled_null`); block-shuffle null and look-elsewhere by simulation `chain/dev_stages.py` (`block_shuffle_order`, `sky_from_z`); null suite `chain/CPG_Null_Runner/cpg_null_runner.py`; per-run tool registry `chain/cmb_tools.py`	masked pseudo angular power spectrum (healpy anafast), six bands; within-chromosome block-shuffle null (runs of 50 sites); look-elsewhere: max band ratio against 100 shuffles of the same array	a HEALPix residual map (stage 11)	development flag `--dev-sky` (2026-10-04). DEV-SKY-02: look-elsewhere rate 91 % on 163 healthy arrays (bar 8.4 %)						
12b	Difference map	`chain/serial_mode.py` (`check_same_person`, `delta_sky`, `delta_cells`, `trajectory`)	per-address beta difference of two draws of one person on the addresses both measured, refused unless both draws share the identifier hash, array type and pipeline; per-cell A differences across draws (design doors/`PROC_SERIAL_01_PREREG.md`)	two runs (bundles and beta vectors) of one person	**wired 2026-10-03** behind `--prior-betas`/`--prior-bundle` (DEV-TOOLKIT-ADDED-01 passed; release check E5): per-address difference + same-person check. Not built: the difference drawn as a sky; `delta_cells/trajectory` (class-era fields) not called						

stage	name (SOP 2b)	module / file	what it does (method)	inputs	status						
5	Met-A per cell (B cells)	`chain/dev_stages.py` (`percell_b`); floor `chain/Runtime Matrices/Met_A_Floors/metA_floors_v1_2_ALLCELLS_development.json`	B-cell Met-A against the composition-matched expectation (atlas v2 parents x stage 2 fractions)	a beta vector, the atlas v2 parquet	development flag `--dev-percell-b` (2026-10-04). DEV-PERCELL-01 and DEV-NEWCELL-01: the new-cell rule is not met (purified 16/22, replicate 0.037)						
8	Self-tare on type II fixed sites	`chain/dev_stages.py` (`anchors`, `selftare_map`, `selftare_ii`); `chain/Runtime Matrices/Development/dev_selftare_typeII_EPIC_v1.json`	affine map of each probe design onto the reference arrays' scale from this array's own low and high fixed-site anchors; Met-A re-read	a beta vector	**wired 2026-10-04** as Stage T step 1 (`chain/conductor_v3.py: stage_t_selftare_ii`, before A and M; DEV-SELFTARE-03); `--dev-selftare-ii` is a no-op alias since. DEV-SELFTARE-02: self-tare II then median tare meets every bar (replicate SD 0.016, 62/63; other laboratories 49/49; floor 6/6) - adopted by the author as the tare on 2026-10-04						

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1	EPIC v2 calibration	`chain/dev_stages.py` (`epic_v2_calibrate`); `chain/dev_epicv2_sesame.R` (SeSAMe)	SeSAMe on an EPIC v2 IDAT pair, v2 probes averaged into EPIC v1 names, read on the v1 floor at the shared sites (cross-version, development)	an EPIC v2 IDAT pair, an Rscript with sesame	development flag `--dev-epic-v2` (2026-10-04). DEV-EPIC-V2-01: 72/72 calibrated; identity-site coverage 5,391 of 6,000; same-blood v2 - v1 SD 0.054; no v2 floor						

Notes

- `CPG_Null_Runner/cpg_null_runner.py` null N7 loads the synthetic patient generator, which was retired with chain v2; N7 reports DEFERRED until a v3 generator exists. The other nulls run without it.
- `cmb_tools.py` checks fields of the retired v2 bundle (`patient_sky`, `foreign_detection`, `second_opinion`); on a v3 bundle every entry reads NOT_RUN until the stage it checks is wired in.
- Kept out, and why (SOP section 2b): the class-era stage 10 panel, 3b panel, 3c panel, 11b archives and the toolkit NILC module N1 (each named in its row above), and null N7 of the null runner (its generator was retired).
- The class-era wrappers that called these modules inside the retired conductor (the stage 4.6 sky wrapper, the class-level NILC second opinion) were not carried over: they read retired files (class identity loci, beta scale maps). Each stage gets a v3 wrapper when it is commissioned.

Sources

manual/OM_v3_neutrophil_chain.md sha256 1f647f23da96

doors/CHAIN_SEQUENCE.md sha256 d654934d4b3a

chain/TOOLKIT.md sha256 d4890053d11f